Room for Carbon? Mapping Short-Term Stays and Emissions

With the rise of Airbnb or short-term rentals, previous hotel goers now use Airbnb as their premium alternative when traveling to new places. Al..

Background

According to a study in Sydney, researchers found that the platform and hosts generate a range of 7.27 to 9.39 kg CO2e per room and night. These findings challenge the notion that short-term rentals contribute to decreased environmental sustainability. We can check for similar results in London.

Airbnb Facts

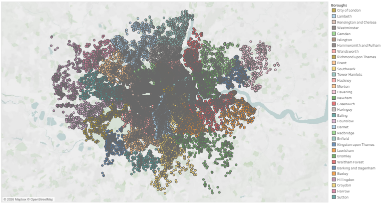
Listings in London
62,684

Per Borough

Average Number of Listings
1,900

Average Carbon Emissions (KiloTonnes)
1,085

Every Airbnb Listing Plotted



Listings appear to be more centralized in the center of London and along the river. Before we connect emissions, let's first find the most expensive boroughs to examine trends between price and number of listings.

TOP 3 Most Expensive London Bouroughs

City of London:
Average Price For Each Rental: $379
Number of Listings: 440

Lambeth:
Average Price For Each Rental: $369
Number of Listings: 2,948

Kensington and Chelsea:
Average Price For Each Rental: $363
Number of Listings: 4,734

The three most expensive boroughs have a varying range of listings. Based on this, there could be a correlation between the price and carbon emissions. City of London lacks listings, but has a high price. Kensington and Chelsea have lots of listings at high prices.

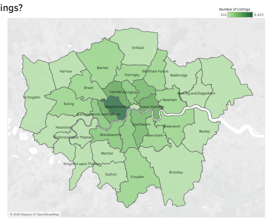
Where are the London Listings?

Boroughs with the most listing

Westminister: 8,423

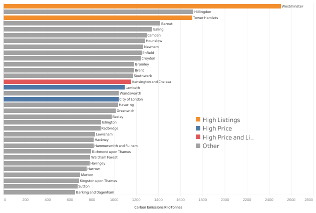
Kensington and Chelsea: 4,734

Tower Hamlets: 4,424



Kensington and Chelsea is the only overlapping borough with lots of listings and a high average listing price. This could help explain that both higher prices and more listings lead to fewer carbon emissions. Westminster, according to the study, should have the highest carbon emissions if the study is representative of London.

Where are Carbon Emissions Concentrated in London?



Most Polluted Boroughs

Emissions measured in Kilotonnes

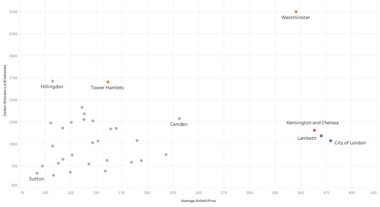
Westminster: 2502
Average Rental Price: **345**
Number of Listings: 8,423

Hillingdon: **1713**
Average Rental Price: **161**
Number of Listings: 706

Tower Hamlets:**1702**
Average Rental Price: **1..**

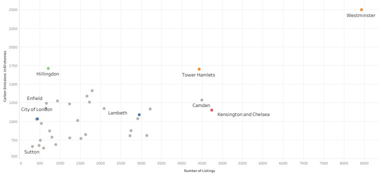
The three most paid for boroughs are not the highest in carbon emissions. People could be paying more in order to have lower carbon emissions. Two of the most densely populated boroughs are the most polluted. This could help prove that the study has legitimacy in London, for increased carbon emissions with more listings. However, to dive more in, we must compare on a more granular level.

Exploring the Connnection Between Price and Emission Level



The most costly boroughs have carbon emissions levels lower than 1200 kilotonnes. Those boroughs could cost more to offset carbon emissions or be the premium for less carbon emissions. One outlier is Westminster, which has a higher average price and high carbon emissions. This makes it harder to find a correlation between price and emissions.

Exploring the Connnection Between Listings and Emission Level



The number of listings does not appear to impact the carbon emissions of a borough. There are boroughs like the City of London and Kensington and Chelsea, with similar emissions but vastly different listings. Westminster, again, is an outlier that helps show that more listings mean mo..

After this analysis, there is no conclusive evidence to say that price accounts for or even impacts the amount of carbon emissions. Westminster sticks out and prevents us from claiming that price helps drive down carbon emissions. There is a lack of evidence that boroughs with more listings have more carbon emissions. To go more in-depth, we could explore which industries dominate each borough to help us understand where the emissions originate.

By Alexander Jung

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